SOFTSOLS PAKISTAN

PJPS Reference Checker

Five-Module Manuscript Screening System

Proposal for Approval

Prepared for: The Editor, Pakistan Journal of Pharmaceutical Sciences (PJPS)

Prepared by: Softsols Pakistan

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Document control

Version	Date	Author	Change
1.0	21 October 2026	Softsols Pakistan	Full platform proposal (all tools)
2.0	24 October 2026	Softsols Pakistan	Scope reduced to the four modules requested by the PJPS Editor; Paper Mill Detection added
2.1	24 October 2026	Softsols Pakistan	Module 5 — Image Integrity Detection added (benchmarked against ImageTwin and Proofig)

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1. Executive summary

Fake and AI-generated references, wrongly cited sources, retracted papers, references that don't follow the journal style, paper-mill manuscripts and duplicated or edited figures now reach PJPS in every submission batch. Checking them by hand takes an editor hours per manuscript, and many problems are only found after publication.

At the Editor's request, this proposal narrows the PJPS Reference Checker to five modules:

#	Module	What it answers
1	Check Bibliography	Is every reference real, correct and not retracted?
2	Reference Style Check & PJPS Converter	Do the references follow PJPS style? If not, convert them.
3	Citations	Do in-text citations match the reference list, and does each source support its claim?
4	Paper Mill Detection	Does this manuscript show signs of coming from a paper mill?
5	Image Integrity Detection	Are any figures duplicated, reused or edited?

A single Full Screening Report runs all five modules on one upload and gives the editor a one-page summary with a desk-review recommendation.

> *Recommendation: build on the existing PJPS Reference Checker instead of starting again. Most of Modules 1–3 already exist and work; all other tools will be switched off and removed from the menu. The new work goes where it adds the most value: the PJPS style compliance check, Paper Mill Detection and Image Integrity Detection.*

2. Approach: modify the existing software

2.1 Why not build a new system

- **Proven core** — the reference engine already checks 12 scholarly databases (Crossref, PubMed, Europe PMC, OpenAlex, Semantic Scholar, DataCite and others) plus Retraction Watch.
- **PJPS style already defined** — the style engine already contains the PJPS house style and converts Word files.
- **Lower cost and risk** — a rebuild would repeat months of matching, database and Word-handling work and bring new defects.
- **Easy to simplify** — the software already has a module/permission system, so other tools can be switched off without removing code, and turned back on later if PJPS wants.

2.2 What changes

Area	Change
Navigation	Only the five modules and the Full Screening Report are shown. Find a Journal, Conferences, Thesis Assistant, Find by Author, Find Evidence, Manuscript Format and Reviewer tools are switched off.
Dashboard	Redesigned around the five modules and the PJPS manuscript number.
New work	PJPS style compliance check, Track Changes output, extra bibliography indicators, merged Citations module, Paper Mill Detection, Image Integrity Detection, Full Screening Report.
Kept	Users and roles, saved reports, retraction alerts, exports, web platform and Windows desktop application.

2.3 Features shared by all five modules

- Upload once (.docx, .doc, .pdf, .odt, .rtf, .txt; .bib/.ris for Module 1) and reuse the file in every module.
- Each check is linked to the PJPS manuscript number, with a history of saved reports.
- Exports: Word, PDF and CSV (plus BibTeX/RIS in Module 1).
- Available on the web and in the Windows desktop application.
- Manuscripts are processed on PJPS-controlled systems; no manuscript text is sent to an outside AI service. The only exception is the optional image vendor check in Module 5, which the editor switches on per manuscript.

3. Module 1 — Check Bibliography

Purpose: confirm that every reference is real, correct and not retracted.

3.1 Input and verification

Feature	Description	Status
Input	Upload a manuscript (the reference list is found automatically) or paste references; import .bib/.ris. The editor can correct the detected list (split or merge entries) before checking.	Existing
Text clean-up	Repairs text broken by PDF extraction (line breaks, hyphenation, special characters) before searching.	Existing
Database check	Crossref, DOI registry, DataCite, OpenAlex, PubMed, Europe PMC, Semantic Scholar, arXiv, DBLP, Open Library, Google Books and ERIC. PubMed and Europe PMC are searched first because PJPS is a biomedical journal.	Existing
Matching	Each reference is matched on title, authors, year, journal, volume, issue, pages and DOI.	Existing

3.2 Results per reference

Feature	Description	Status
Verdict	Verified (green), Miscited (yellow: real work, wrong details), Unverified (grey), Suspicious (orange), Retracted (red).	Existing
Confidence score	0–100 score for each reference.	Existing
Comparison	The matched record is shown next to the original, with the differing fields highlighted.	Existing
Corrected reference	The correct reference in PJPS style, ready to copy.	Existing
Source link	Link to the matched record (DOI, PubMed or publisher).	Existing

3.3 Fake and AI-generated reference checks

Feature	Description	Status
DOI checks	DOI doesn’t resolve, or points to a different paper; DOIs with page numbers or junk attached are repaired.	Existing
PMID / arXiv checks	Identifiers that don’t exist or point to another work.	Existing
Journal mismatch	Journal name doesn’t match the DOI or the database record.	Existing
Altered titles	Real titles with invented words inserted.	Existing
Unknown authors	Author names not found on the matched work.	Existing
Missing from indexes	A reference that claims to be in an indexed journal but is in no database is marked Suspicious.	Existing
Fewer false alarms	Multi-line references, book reviews, thesis-based articles and Pakistani or regional journals that aren’t indexed are shown as “Not indexed”, not as fake.	Improve

3.4 Retractions, links and reference age

Feature	Description	Status
Retractions	Retraction Watch: retraction, expression of concern or correction, with reason and date.	Existing
Retraction monitoring	The editor is alerted if a reference in a saved report is retracted later.	Existing
Link check	Every URL and DOI link is tested (working, broken or redirected), separately from the verdict, because a dead link doesn't mean a fake reference.	Existing
Reference age	Chart of references by year; share from the last 5 and 10 years; list of references older than 10 years; warning when recent references fall below a limit set by PJPS.	Improve

3.5 Extra quality indicators

Feature	Description	Status
Self-citation	References that include the manuscript's own authors.	New
PJPS citations	References to PJPS articles, counted separately.	New
Predatory / hijacked journals	References to journals on predatory or hijacked journal lists.	New
Duplicates	The same work listed more than once.	New

3.6 Output

- Summary bar with counts per verdict; filter by verdict.
- Exports: Word report, PDF, CSV, BibTeX, RIS and a corrected reference list in PJPS style.
- Re-check button that shows only what changed since the last run.

4. Module 2 — Reference Style Check & PJPS Converter

Purpose: check that references and in-text citations follow PJPS style, and convert them when they don't.

4.1 PJPS house style

Element	PJPS format
In-text, one author	(Khan, 2019)
In-text, two authors	(Khan and Ali, 2019)
In-text, three or more	(Khan et al., 2019)
Journal article	Khan A, Ali B and Shah C (2019). Title of article. Pak. J. Pharm. Sci., 32(1): 1-5.
Book	Khan A and Ali B (2019). Title of book. Publisher, Place.
Order	Alphabetical by first author, then year; 2019a / 2019b for the same author and year.

To be confirmed by the PJPS Editor: these rules come from the house style already in the software. The compliance check will enforce exactly the rules PJPS approves, so please check them against the current Instructions to Authors.

4.2 Style compliance check

Feature	Description	Status
Author format	Surname followed by initials without full stops; commas between authors; "and" before the last author; the number of authors before "et al.".	New
Year	Year in parentheses after the authors.	New
Title	Sentence case, ending with a full stop.	New
Journal name	Correct NLM/ISO abbreviation.	New
Volume, issue, pages	Format Vol(Issue): pages; page range with a hyphen; no "pp.".	New
Missing details	Missing volume, issue, pages, year or publisher.	New
Order	Alphabetical order; correct a/b suffixes.	New
In-text format	(Author, Year) format, correct "et al." use, semicolons between citations, chronological order inside one bracket.	New
Result	Percentage of references that comply, plus each problem with the expected correct form.	New

4.3 Converter to PJPS format

Feature	Description	Status
Input	Word manuscript or a pasted reference list.	Existing
Source styles	APA, Vancouver / numbered [1], Harvard, IEEE, AMA, Chicago, Nature, Elsevier and others, detected automatically.	Existing
Full conversion	Converts both the in-text citations and the reference list; numbered [1] becomes (Author, Year) and the list is re-sorted alphabetically.	Existing
Completing references	Missing DOI, volume, issue and pages are filled from the matched record (Module 1 engine).	Existing
No silent changes	References that can't be matched, and suspicious DOIs, are flagged for the editor instead of being changed.	Existing
Layout kept	Only citations and references change; the rest of the manuscript layout stays	Existing

Feature	Description	Status
	the same.	

4.4 Review and output

Feature	Description	Status
Before / after view	Original and converted references side by side.	Existing
Track Changes	Changes appear in the Word file as Track Changes, so the editor or author can accept or reject each one.	New
Downloads	Converted .docx, a copyable reference list, and a report of what changed and what still needs a manual fix.	Improve

5. Module 3 — Citations

Purpose: check that in-text citations and the reference list match, and that each cited source really supports the claim made.

5.1 Citations vs reference list

Feature	Description	Status
Missing references	Cited in the text but not in the reference list.	Existing
Uncited references	In the reference list but never cited.	Existing
Mismatches	Different year or author spelling between the text and the list.	Existing
Ambiguous citations	“Khan, 2020” when the list has 2020a and 2020b, or two different authors named Khan in 2020.	Existing
“et al.” misuse	For example “et al.” used for a two-author work.	New
Numbered citations	Gaps, numbers out of sequence, numbers with no matching reference.	Existing
Whole document	Citations in tables, figure captions and footnotes are included; each problem links to its location.	Improve

5.2 Claim support

Feature	Description	Status
Claim extraction	The text is split into cited claims (a sentence with its citation).	Existing
Comparison	Each claim is compared with the cited source: an uploaded PDF (with page number), open-access full text, or the abstract.	Existing
Verdicts	Full support, Partial, Unsupported or Could not verify, with the supporting passage from the source.	Existing
PDF matching	Shows which uploaded PDF matched which reference, and separates “not found in your PDF” from “source not accessible”.	Existing
Retracted evidence	A claim supported by a retracted paper is flagged red.	Existing
Privacy	Runs on PJPS systems; manuscript text isn’t sent to an outside AI.	Existing

5.3 Citation patterns

Feature	Description	Status
Self-citation	Heavy citation of the authors’ own work.	Existing
Citation stacking	Many citations to one journal or one group of authors.	Existing
Citation blocks	Large groups of citations with no specific claim (for example [1–25]).	New

5.4 Output

- One Citations report combining list matching, claim support and pattern flags.
- Exports: Word, PDF and CSV.

- Author letter in English or Urdu listing the citation problems to fix. The editor previews and approves it before it is sent.

6. Module 4 — Paper Mill Detection

A paper mill is a business that sells fake or template-based research papers and authorship places, often for degree or promotion requirements. Pharmaceutical and biomedical journals are among the most targeted. Typical examples are template "drug X inhibits cancer via the A/B pathway" papers, network-pharmacology and molecular-docking papers, and studies with recycled images or invented data.

Purpose: alert the editor to signs that a manuscript may come from a paper mill. This module is for editors only.

> *Module 4 is a screening aid, not a judgement. It gives a LOW / MEDIUM / HIGH risk level and the evidence for each signal, so the editor knows where to look closer. It never rejects a paper by itself, and no tool can detect paper mills with certainty. The final decision always stays with the editor.*

6.1 Text signals

Feature	Description	Status
Tortured phrases	Odd synonyms used to get past plagiarism checks, such as "counterfeit consciousness" for artificial intelligence or "bosom peril" for breast cancer. Uses the published Problematic Paper Screener list plus a PJPS pharmacy list the editors can add to.	Phase 1
AI-text leftovers	Clear traces such as "As an AI language model", "Regenerate response" or placeholder text. Only definite matches are flagged; there is no AI-probability score, because such scores are unreliable.	Phase 1
Template wording	Wording typical of mill templates (network pharmacology, docking, "miR-X sponges Y" papers).	Phase 2
Vague methods	Missing supplier, catalogue number, dose, sample size or ethics approval number.	Phase 2

6.2 Reference signals (using Module 1)

Feature	Description	Status
Fake references	Results of the Module 1 fake-reference checks.	Phase 1
Paper-mill retractions	References to papers retracted by publishers for paper-mill reasons (Retraction Watch).	Phase 1
Problem journals	Many references to predatory, hijacked or mill-affected journals.	Phase 1
Citation stacking	Results of the Module 3 citation pattern checks.	Phase 1

6.3 Author and submission signals

Feature	Description	Status
Earlier retractions	Authors with earlier retractions, especially for paper mills or fake peer review (by name and ORCID).	Phase 1
E-mail addresses	Free e-mail addresses instead of institutional ones.	Phase 1
ORCID	Missing ORCID, or a new ORCID with no publications.	Phase 1
Affiliations	Affiliations that don't fit the topic, or distant institutions with no clear collaboration.	Phase 2

Feature	Description	Status
Authorship changes	Authors added, removed or reordered between the original and the revised version.	Phase 2
Suggested reviewers	Suggested reviewers with free e-mail addresses or links to the authors.	Phase 2

6.4 Data and biology signals

Feature	Description	Status
Cell lines	Misidentified or contaminated cell lines named in the methods (Cellosaurus / ICLAC register).	Phase 2
Duplicate submission	Comparison with earlier PJPS submissions to find the same text with small edits or a recycled title and abstract (figures through the Module 5 image fingerprint store).	Phase 2
Primer and gene sequences	Sequences checked against NCBI to find wrong or impossible primers.	Phase 3
Statistics	Percentages, sample sizes or p-values that can’t come from the stated data.	Phase 3
Images	Duplicated or edited images are now covered by Module 5; its findings count as one signal in the paper mill risk level.	Module 5

6.5 Output

- Risk badge (LOW / MEDIUM / HIGH) with each signal, its weight and its evidence (quoted text, matched record, link).
- Editor notes and a decision record: cleared, needs investigation, or referred to the integrity committee.
- Confidential PDF/Word export for the editorial file.
- Never shown to authors or reviewers.

7. Module 5 — Image Integrity Detection

Duplicated, reused and edited figures (Western blots, gels, microscopy, histology) are among the most common reasons for retraction in biomedical journals, and a typical sign of paper-mill work. Commercial tools such as ImageTwin and Proofig check for them; this module brings the same kind of check into the PJPS system.

Purpose: find figures that are duplicated, reused or edited, and show the editor the evidence side by side. This module is for editors only.

> *Module 5 is a screening aid, not a judgement. Repeated textures such as cells, tissue or gel background can look like copies, so every finding is shown as images side by side for the editor to confirm or dismiss. By default no images leave PJPS systems.*

7.1 Comparison with commercial tools

We reviewed ImageTwin and Proofig (October 2026). The table shows which of their features this module covers.

Feature	ImageTwin / Proofig	PJPS Module 5
Duplication inside the manuscript (rotated, flipped, resized, partly overlapping)	Yes	Yes — built in (Phase 1)
Cloned areas inside one image	Yes	Yes — built in (Phase 1)
Splicing and other edits	Yes	Forensic viewer for the editor (Phase 1)
Western blot / gel lane checks	Yes	Yes — built in (Phase 2)
Reuse of earlier PJPS figures	No	Yes — PJPS image store (Phase 2)
Reuse of the authors' own earlier figures	Proofig	Open-access papers via Europe PMC (Phase 2)
Comparison against 155–160 million published figures	Yes	Through the optional vendor connection
AI-generated image detection	Yes	Through the optional vendor connection
Report	Yes	Yes, plus editor decisions and author query letter

> *The two features marked "vendor connection" need a database of over 150 million figures and AI models trained on proprietary data. They can't be rebuilt reliably in-house, so the module offers them through the ImageTwin or Proofig API instead. Without that connection they are reported as "not checked".*

7.2 Image extraction

Feature	Description	Status
Extraction	Images are taken out of .docx and .pdf files at full resolution; separate figure files (TIFF, PNG, JPG) are also accepted.	Phase 1
Panels	Multi-panel figures are split into sub-panels; panel labels and figure numbers are read, so each finding says, for example, "Fig. 2C vs Fig. 4A".	Phase 1
Figure type	Each image is classed as Western blot / gel, microscopy / histology, photo or graph.	Phase 1

7.3 Detection

Feature	Description	Status
Duplicated images	The same image or panel used twice, including copies that are rotated, flipped, resized, cropped or only partly overlapping. Matched areas are outlined on both images.	Phase 1
Cloned areas	Part of an image copied to another place in the same image. Shown as “check this area”, not as a verdict.	Phase 1
Forensic viewer	Contrast boost, error level analysis, noise map, edge view and an overlay slider, to help the editor spot splicing, erased bands or pasted backgrounds.	Phase 1
Western blots and gels	Lanes and bands compared with each other: repeated, flipped or stretched lanes, sharp cut lines, and the same band used for different proteins.	Phase 2
Earlier PJPS figures	Fingerprints of every PJPS submission and published paper are stored; each new manuscript is compared with all of them.	Phase 2
Authors’ earlier figures	Figures from the authors’ earlier open-access papers (found through Module 1 and ORCID) are downloaded from Europe PMC and compared.	Phase 2

7.4 Optional vendor connection

Feature	Description	Status
Published-figure check	Comparison against 155–160 million published figures through the ImageTwin or Proofig API.	Optional
AI-generated images	Detection of figures made with image-generating AI, through the same API.	Optional
Control	Off by default; the editor switches it on per manuscript, because the figures are sent to the vendor. The vendor charges PJPS per use (Proofig’s published prices start at USD 99 for 120 sub-images; ImageTwin quotes on request).	Optional

7.5 Output

- Image risk badge (LOW / MEDIUM / HIGH) and a list of findings, each with the two images side by side, the matched area outlined, figure and panel labels, and a confidence level.
- The editor marks each finding as confirmed, false alarm or ask the authors; the decision is kept with the manuscript.
- “Ask the authors” prepares a query letter with the image pairs, asking for the original images or raw blots. The editor approves it before it is sent.
- Confidential PDF/Word export for the editorial file.
- Findings count as one signal in the Module 4 paper mill risk level.

7.6 Technical notes

- Runs in the background on the existing job queue; about 1–3 minutes for a manuscript with around 30 figures.
- In the Windows desktop application, extraction, detection and the forensic viewer work offline; the Europe PMC and vendor checks run when the connection is available.
- Uses established open-source image libraries (OpenCV, MuPDF, Tesseract OCR); no new paid software is needed for the in-house checks.

8. Full Screening Report

One upload runs all five modules and produces a single report.

Section	Summary shown on the front page
Bibliography	Verified, miscited, suspicious and retracted counts
Style	Percentage of references that comply with PJPS style
Citations	Missing, uncited and unsupported citations
Paper mill risk	LOW / MEDIUM / HIGH
Image integrity	Number of findings to check; LOW / MEDIUM / HIGH
Recommendation	Proceed to review, return to author, or investigate

- **Editor version** — all five modules, including Paper Mill Detection and Image Integrity Detection.
- **Author version** — Modules 1–3 only; the editor previews and approves it before it is sent.

9. Users and access

Role	Access
Super Admin	Creates and manages all user accounts, roles and module permissions; manages datasets and settings. There is no public sign-up.
Editor	All five modules and the Full Screening Report.
Reviewer	Modules 1–3 if the Super Admin enables them. No access to Paper Mill Detection or Image Integrity Detection.

10. Implementation plan

The timeline below is indicative and will be confirmed after approval.

Stage	Work	Indicative time
1. Set-up	Switch off the other tools, new menu and dashboard, manuscript number field	Week 1
2. Module 1	Fewer false alarms, reference age limits, new quality indicators	Weeks 2–3
3. Module 2	PJPS style compliance check, Track Changes output, change report	Weeks 3–5
4. Module 3	Merged Citations module, “et al.” checks, citation blocks, whole-document coverage	Weeks 5–6
5. Module 4, phase 1	Tortured phrases, AI-text leftovers, reference and author signals, risk report	Weeks 6–8
6. Module 5, phase 1	Image extraction, duplicated images, cloned areas, forensic viewer, image report	Weeks 8–11
7. Full report and testing	Full Screening Report, testing with sample PJPS manuscripts, training, go-live	Weeks 11–12
8. Module 4, phases 2–3	Template wording, cell lines, duplicate submissions, authorship changes; then sequences and statistics	After go-live, agreed separately
9. Module 5, phase 2 and vendor option	Western blot / gel checks, PJPS image store, authors’ earlier figures; optional ImageTwin or Proofig connection	After go-live, agreed separately

10.1 Deliverables

- Updated web platform and Windows desktop application with the five modules.
- Editor and Super Admin user guides.
- Test report using sample PJPS manuscripts and an acceptance sign-off sheet.
- Training session for the editorial team.

10.2 Acceptance criteria

1. Each of the five modules runs end to end on sample PJPS manuscripts, on the web and on the desktop.
2. The style compliance check enforces the PJPS rules approved by the Editor.
3. Converted Word files keep their layout and show changes as Track Changes.
4. Paper Mill Detection is visible only to editors and shows the evidence for every signal.
5. Image Integrity Detection finds duplicated panels (including rotated, flipped and resized copies) in a test set of manuscripts with known problems, and shows every finding side by side.
6. The Full Screening Report produces both an editor version and an author version.

11. Decisions needed from the Editor

#	Decision	Our suggestion
1	Approve the five modules and the Full Screening Report as the final scope	Approve
2	Approve modifying the existing software instead of building a new system	Modify
3	Confirm the PJPS reference and in-text citation rules (Section 4.1)	Check against the current Instructions to Authors
4	Set the minimum share of recent references (last 5 years)	For example 30%
5	Approve Module 4 phase 1 now, with phases 2–3 to follow	Phase 1 now
6	Who receives the author version of reports	Corresponding author, after editor approval
7	Allow reviewers to use Modules 1–3	Optional, set by the Super Admin
8	Approve Module 5 phase 1 now, with phase 2 to follow	Phase 1 now
9	Whether to subscribe to an image vendor (ImageTwin or Proofig) for the published-figure and AI-image checks	Decide after phase 1; ask both for a journal quote
10	Provide sample manuscripts with known image problems for testing	Retracted or corrected PJPS papers, if available

12. Assumptions and limitations

12.1 Assumptions

- PJPS continues to provide hosting and the PostgreSQL database used by the current system.
- PJPS provides a sample set of manuscripts (with personal data removed) for testing.
- External databases (Crossref, PubMed, OpenAlex, Retraction Watch and others) stay publicly available under their current terms; their availability is outside Softsols Pakistan's control.
- Commercial terms are provided separately. Any ImageTwin or Proofig subscription is contracted and paid by PJPS directly with the vendor.

12.2 Limitations

- All results support the editor's decision; they don't replace it. A green result doesn't guarantee a reference is genuine, and a red one doesn't prove fabrication.
- Claim-support results depend on access to the source; abstract-only checks are labelled as such.
- Paper Mill Detection shows risk signals, not proof. No tool detects paper mills with certainty.
- Image Integrity Detection can miss edits hidden by heavy compression, and repeated textures can cause false alarms. Without the vendor connection it doesn't compare figures with the wider published literature or detect AI-generated images.

12.3 Approval

By signing below, PJPS approves the scope described in this proposal.

	Pakistan Journal of Pharmaceutical Sciences	Softsols Pakistan
Name		
Designation		
Signature		
Date		